

Metalwork Basic Hand, Marking and Cutting Tools

1. Steel Rule



2. Measuring Tape



3. Center Punch



4. Scriber



5. Steel Square



6. Files



7. Combination Set



8. Hacksaw



9. Surface Gauge



11. Mallet



13. Bevel Square



15. Vernier Caliper



10. Tin-Snip



12. Hammers



13. Divider



16. Vice



Video : <https://youtu.be/kqxcgzpobygM>

Feedback : Activity

Question : 1

1. Name The Following Basic Hand, Marking and Cutting Tools.



1. _____



2. _____



3. _____



4. _____



5. _____



6. _____



7. _____



8. _____



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10. _____



11. _____



12. _____

Measure With a Steel Rule And Mark Off With A Scriber on Tubing



Step 1 : Spray the workpiece with the Engineer's Blue or use a permanent marker.

Step 2 : Take a Steel Ruler and measure the required length.

Step 3 : Use a Scriber and the Engineer's Square to draw a straight line.

Step 4 : Double check if the measurement is correct.

Step 5 : Use the center punch to make pop marks on the line

Video :

- <https://youtu.be/V2QT2C5larE>

Feedback : Practical Activity

Cut a Small Bar or Tube With a Hacksaw or Tinsnip



Step 1 : Clamp the steel bar in a vise.

Step 2 : Measure the required length on the workpiece.

Step 3 : Use a hacksaw frame with a high-quality blade to start cutting.

Step 4 : Apply firm, steady pressure on the down stroke of the saw.

Step 5 : Maintain sawing motion until the cutting is completed.

Video :

- <https://youtu.be/HVh5sRxnUbU>

Feedback : Practical Activity

File To Square With A Hand File



Step 1 : Decide on master edge, file it roughly flat.

Step 2 : Hold it against the Try Square to see how flat it is.

Step 3 : Try to eliminate the high spots using a file.

Step 4 : Recheck the edge flatness with a Try Square.

Step 5 : Then start doing the same process with the adjacent ends, using the Try Square taking data from the 'master edge'.

Step 6 : Finally wrap up the opposite side of the master edge using one of the side as the datum for the Try Square.

Video :

- <https://youtu.be/BM8gZuLr0CE>

Feedback : Practical Activity